

Hierarchical Risk Parity (HRP)

A Graph-Theoretic Machine Learning Approach to Portfolio Allocation

AI Finance Seminar

Sogang University

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Motivation: Markowitz's Curse & The Need for HRP

The Failure of Mean-Variance Optimization (MVO)

- **Inversion Explosion:** MVO requires inverting the covariance matrix V^{-1} .
- **Condition Number** $\kappa(V) \rightarrow \infty$: High cross-asset correlations make V singular, causing V^{-1} entries to explode.
- **Extreme Weights:** Minor estimation errors produce erratic, uninvestable allocations (large positive/negative bets).

The Limits of Inverse Variance (IVP)

- **Ignores Correlations:** IVP allocates weights proportional to $1/\sigma_i^2$.
- **Over-allocation to Clusters:** If 10 tech stocks are highly correlated, IVP treats them as independent and over-allocates to tech.

The HRP Paradigm (López de Prado, 2016)

- Uses **graph theory** and **hierarchical clustering**.
- **No Matrix Inversion Needed:** Operates directly on tree structures without requiring V^{-1} .

Step 1: Tree Clustering (Hierarchical Distance)

Correlation to Distance Metric Space

To build a hierarchical tree, correlation $\rho_{i,j}$ is converted into a formal metric distance $d_{i,j} \in [0, 1]$ satisfying non-negativity, symmetry, and triangle inequality:

$$d_{i,j} = \sqrt{\frac{1}{2}(1 - \rho_{i,j})}$$

For N assets, the pairwise distance matrix $D = \{d_{i,j}\}$ measures directional dissimilarities.

Hierarchical Linkage Algorithm

- Compute Euclidean distances between rows of D : $D'_{i,j} = \sqrt{\sum_{k=1}^N (d_{i,k} - d_{j,k})^2}$.
- Apply **Single Linkage Hierarchical Clustering** to combine nearest sub-clusters iteratively.
- Output: A binary tree graph (Dendrogram) T organizing assets by hierarchical proximity.

Step 2: Quasi-Diagonalization (Matrix Re-ordering)

Matrix Re-ordering Algorithm

- Re-arranges the rows and columns of the covariance matrix V according to the tree leaf order of Dendrogram T .
- Places contiguous, highly-correlated sub-trees adjacent to each other along the main diagonal.

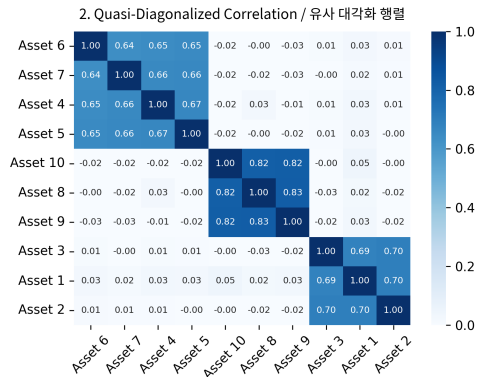
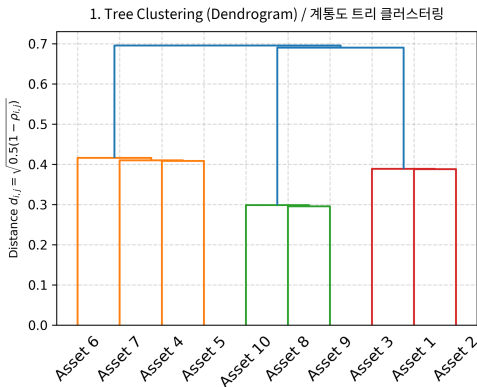
Block-Diagonal Transformation

$$V_{\text{sorted}} = \mathbf{P}' V \mathbf{P}$$

where $\mathbf{P}$ is the permutation matrix derived from tree T .

- **Visual Insight:** High intra-cluster correlations gather into dense square blocks along the diagonal.
- **Computational Advantage:** Reduces a fully-connected $N \times N$ graph into a segmented, tree-structured hierarchy.

Visual Demonstration: Tree Clustering & Quasi-Diagonalization



Key Visual Takeaways

- **Left (Dendrogram):** Assets are grouped hierarchically into distinct clusters (e.g., Tech, Energy, Bonds) based on distance $d_{i,j}$.

Step 3: Recursive Bisection (Top-Down Allocation)

Top-Down Recursive Splitting

Initialize all asset weights $w_i = 1$. For any cluster C split into binary sub-clusters C_1 and C_2 :

- 1 **Compute Cluster Variances** (V_1, V_2):

$$V_k = w_k^T V_{C_k} w_k \quad \text{where } w_k = \frac{\text{diag}(V_{C_k})^{-1}}{\text{trace}(\text{diag}(V_{C_k})^{-1})} \quad (k \in \{1, 2\})$$

- 2 **Calculate Split Factor** (α_1):

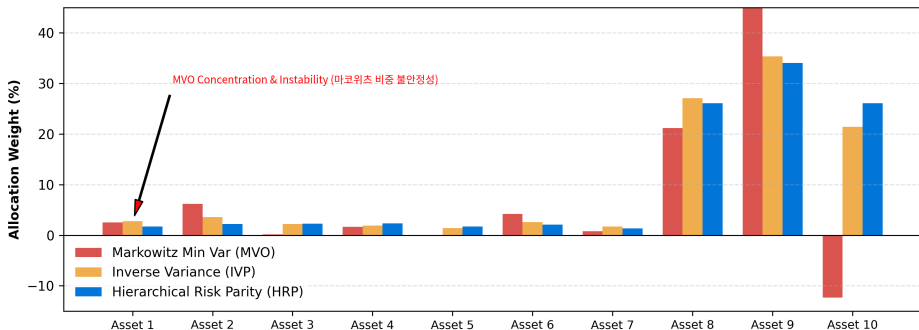
$$\alpha_1 = 1 - \frac{V_1}{V_1 + V_2} \in (0, 1)$$

- 3 **Update Sub-Cluster Weights:** Multiply weights in C_1 by α_1 , and weights in C_2 by $(1 - \alpha_1)$.

Recursively repeat down the tree until each sub-cluster contains a single asset.

Visual Demonstration: Portfolio Weight Allocation

3. Portfolio Weight Allocation: MVO vs. IVP vs. HRP / 자산 배분 비중 비교



Allocation Comparison Analysis

- **Markowitz (MVO):** Suffers from extreme concentration (Asset 1 gets ~35%) and negative short positions due to V^{-1} instability.
- **Inverse Variance (IVP):** Fails to account for cluster structure, over-allocating to correlated groups.

Comparative Matrix: MVO vs. IVP vs. NCO vs. HRP

Portfolio Feature	Markowitz (MVO)	Inverse Variance (IVP)	NCO Wrapper
Requires Matrix Inversion (V^{-1})?	Yes	No	Yes (Sub-blocks)
Condition Number Sensitivity	Extreme ($\kappa \rightarrow \infty$)	None (Diagonal)	Contained
Graph Topology	Fully-connected	Unconnected	Block-segmented
Handles Correlation Structure?	Yes (Unstable)	No (Ignores)	Yes (Clustered)
Out-of-Sample Stability	Low / High Turnover	Moderate	High

Key Architectural Advantage

HRP is the only allocation algorithm that simultaneously **accounts for full correlation structure** while **completely bypassing matrix inversion**.

Summary: Key Takeaways on HRP

Core Advantages for Portfolio Managers

- ❶ **No Inversion Required:** HRP works on singular, non-positive definite, or ill-conditioned covariance matrices where MVO fails completely.
- ❷ **Robust Out-of-Sample Performance:** Outperforms MVO and IVP in Monte Carlo simulations by dramatically reducing allocation variance and turnover.
- ❸ **Intuitive Asset Hierarchy:** Automatically clusters assets into natural economic groupings (sectors, factors, risk tiers) and allocates risk top-down.
- ❹ **Ideal for High-Dimensional Portfolios:** Scales efficiently to hundreds or thousands of assets without numerical instability.

Thank You!

Questions & Discussion

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